Find the Error

Grade 12 · S.ID.3

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A data set has a mean of 10. An outlier of 100 is added to the high end. Does the mean increase?

Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The outlier raises the total, so the mean increases.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Which measure of center is resistant to outliers: mean or median? Student's answer: mean

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The median depends only on the middle values, so it is resistant. The mean is pulled by outliers.**

Correct answer: **Median**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

The data 1 through 9 gets an outlier of 100 added. What is the new range? Student's answer: 100

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The range is 100 - 1 = 99. The student gave the maximum.**

Correct answer: 99

Find the Error

Grade 12 · S.ID.4

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A normal distribution has mean 100 and standard deviation 15. About 68% of values lie between which two numbers? Student's answer: 70 and 130

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **70 to 130 is within 2 standard deviations, about 95%. Within 1 standard deviation, the values are 85 and 115.**

Correct answer: **85 and 115**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

With mean 100 and standard deviation 15, what is the z-score of 130? Student's answer: 30

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not divide by the standard deviation. z = (130 - 100) ÷ 15 = 2.**

Correct answer: 2

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a normal distribution, about what percent of values lie more than 1 standard deviation above the mean? Student's answer: 50%

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Half of values are above the mean, but only about 16% are more than 1 standard deviation above it.**

Correct answer: **About 16%**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A survey of 100 students is taken to learn about the whole school. What is the whole school?

Student's answer: the sample

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The 100 students are the sample. The whole school is the population.**

Correct answer: **The population**

Find the Error

Grade 12 · S.IC.1

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

The mean of all voters' opinions is a parameter. Is it a parameter or a statistic? Student's answer: statistic

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The mean of all voters describes the whole population, so it is a parameter.**

Correct answer: **Parameter**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A statistic is a number computed from which group? Student's answer: the population

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A statistic is computed from a sample. A parameter is computed from the population.**

Correct answer: **A sample**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A model says a coin lands heads 10% of the time. In 50 flips, there are 20 heads. Is the model plausible? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The model predicts about 5 heads in 50 flips, but 20 is far from that. The model is not plausible.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A model says a die is fair. In 60 rolls, a 6 comes up 30 times. Is the model plausible? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **A fair die gives 6 about 10 times in 60 rolls. 30 is far from the prediction, so the model is not plausible.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A model says 50% of students like pizza. In a sample of 20, 11 like pizza. Is the model plausible?

Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The model predicts about 10 of 20, and 11 is close. The model is plausible.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A survey uses only people who choose to answer an online poll. Is this sample biased? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **People who choose to respond may differ from the population, so the sample is biased.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A survey asks: Don't you agree that homework is awful? Is the question biased? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The wording pushes respondents toward one answer, so the question is biased.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A random sample of 100 is selected from a population. Is the sample mean an unbiased estimate of the population mean? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Random sampling gives an unbiased estimate, so the sample mean is unbiased.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A random sample of 200 people has 120 who support a plan. Estimate the population proportion. Student's answer: 120

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave the count. The proportion is $\frac{120}{200}$ = 60%.**

Correct answer: **60%**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A poll has a sample proportion of 60% and a margin of error of ±3%. What is the interval?

Student's answer: 3% to 63%

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted incorrectly. The interval is 57% to 63%.**

Correct answer: **57% to 63%**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A sample mean is 40 with a margin of error of ±2. What is the interval? Student's answer: 40 to 42

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only to the mean. The interval is 38 to 42.**

Correct answer: **38 to 42**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Why are subjects randomly assigned to treatment and control groups? Student's answer: to make the groups different

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Random assignment balances other factors between the groups, so differences can be attributed to the treatment.**

Correct answer: **To balance other factors between groups**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A randomized experiment finds a treatment causes improvement. Can we conclude the treatment caused the change? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Randomized experiments allow causal conclusions when the difference is significant.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Students who eat breakfast have higher grades in an observational study. Can we conclude breakfast causes higher grades? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Observational studies show association, not cause. Other factors may explain the difference.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An observational study finds coffee drinkers have higher risk. A headline says coffee causes the risk. Is this conclusion supported? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Observational studies show association, not cause, so the headline overstates the finding.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A survey of 50 teachers in one school is reported as showing what all teachers think. Is this a valid generalization? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Fifty teachers from one school do not represent all teachers, so the generalization is not valid.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A random sample of 1,000 or a random sample of 10 gives an estimate. Which is more reliable?

Student's answer: 10

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Larger random samples give more reliable estimates. The sample of 1,000 is more reliable.**

Correct answer: **1,000**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

How many outcomes are in the sample space for flipping two coins? Student's answer: 2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The outcomes are HH, HT, TH, and TT, so there are 4.**

Correct answer: 4

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

How many outcomes are in the event 'at least one head' when flipping two coins? Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The outcomes are HH, HT, and TH, so there are 3.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

On a standard die, what is the complement of the event 'rolling an even number'? Student's answer: {2, 4, 6}

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student listed the even numbers again. The complement is {1, 3, 5}.**

Correct answer: **{1, 3, 5}**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Are the events 'first flip is heads' and 'second flip is heads' independent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The outcome of the first flip does not change the second flip, so the events are independent.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. Are A and B independent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **0.5 × 0.4 = 0.2, so the events are independent.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A) = 0.3$, $P(B) = 0.6$, and $P(A \text{ and } B) = 0.3$. Are A and B independent? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **0.3 × 0.6 = 0.18, which is not 0.3. The events are not independent.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

P(A and B) = 0.2 and P(B) = 0.5. What is P(A | B)? Student's answer: 0.2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student repeated P(A and B). P(A | B) = 0.2 ÷ 0.5 = 0.4.**

Correct answer: **0.4**

Find the Error

Grade 12 · S.CP.3

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A class has 18 girls and 12 boys. 6 girls wear glasses. What is P(glasses | girl)? Student's answer: $\frac{1}{5}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used all 30 students. The condition is a girl, so P = $\frac{6}{18} = \frac{1}{3}$.**

Correct answer: $\frac{1}{3}$

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A) = 0.4$ and $P(A \mid B) = 0.4$. Are A and B independent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **$P(A \mid B) = P(A)$, so the events are independent.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

In a survey of 100 people, 40 have trait A, 30 have trait B, and 15 have both. What is P(A or B)?

Student's answer: 0.70

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not subtract the overlap. P(A or B) = (40 + 30 − 15) ÷ 100 = 0.55.**

Correct answer: **0.55**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Using the same survey, what is P(A and B)? Student's answer: 0.40

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the count for trait A. Both traits are 15 out of 100, so P = 0.15.**

Correct answer: 0.15

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Using the same survey, what is P(B | A)? Student's answer: 0.15

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave P(A and B). P(B | A) = 15 ÷ 40 = 0.375.**

Correct answer: 0.375

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A) = 0.2$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.1$. Are A and B independent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **$0.2 \times 0.5 = 0.1$, so the events are independent.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Events A and B are mutually exclusive, with $P(A) = 0.3$ and $P(B) = 0.4$. Are they independent?

Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Mutually exclusive events have P(A and B) = 0, but 0.3 × 0.4 = 0.12. They are not independent.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

If $P(A \mid B) = P(A)$, are A and B independent? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Independence means knowing B does not change the probability of A, which is exactly $P(A \mid B) = P(A)$.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A \text{ and } B) = 0.05$ and $P(B) = 0.25$. What is $P(A \mid B)$? Student's answer: 0.05

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student repeated P(A and B). P(A | B) = 0.05 ÷ 0.25 = 0.2.**

Correct answer: **0.2**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A box has 50 items and 5 are defective. If the first item is defective and not replaced, what is P(second is defective)? Student's answer: $\frac{5}{50}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the original probability. After one defective item is removed, 4 defective items remain out of 49, so the probability is $\frac{4}{49}$.**

Correct answer: $\frac{4}{49}$

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

There are 40 seniors, and 15 of them are in honors. What is P(honors | senior)? Student's answer: $\frac{15}{60}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 60 instead of 40. P = $\frac{15}{40} = \frac{3}{8}$.**

Correct answer: $\frac{3}{8}$

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$? Student's answer: 0.9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not subtract the overlap. $P(A \text{ or } B) = 0.5 + 0.4 - 0.2 = 0.7$.**

Correct answer: 0.7

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A and B are mutually exclusive, with $P(A) = 0.2$ and $P(B) = 0.3$. What is $P(A \text{ or } B)$? Student's answer: 0.06

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied. For mutually exclusive events, P(A or B) = 0.2 + 0.3 = 0.5.**

Correct answer: **0.5**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

P(A) = 0.6, P(B) = 0.5, and P(A and B) = 0.3. What is P(A or B)? Student's answer: 1.1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not subtract the overlap. P(A or B) = 0.6 + 0.5 − 0.3 = 0.8.**

Correct answer: 0.8

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A game pays \$10 with probability $\frac{1}{5}$ and costs \$2 with probability $\frac{4}{5}$. What is the expected value? Student's answer: 3.6

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the loss instead of subtracting it. Expected value = $10(\frac{1}{5}) - 2(\frac{4}{5}) = 2 - 1.6 = 0.4$.**

Correct answer: **0.4**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the expected value of rolling a fair six-sided die? Student's answer: 3

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The average of 1 through 6 is 21 ÷ 6 = 3.5.**

Correct answer: 3.5

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A random variable X takes values 0, 1, and 2 with probabilities 0.5, 0.3, and 0.2. What is $E(X)$?

Student's answer: 1

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **$E(X) = 0(0.5) + 1(0.3) + 2(0.2) = 0.7$.**

Correct answer: **0.7**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

X is 0 with probability 0.5 and 2 with probability 0.5. What is the variance of X? Student's answer: 2

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **$E(X) = 1$ and $E(X^2) = 2$. Variance $= 2 - 1^2 = 1$.**

Correct answer: **1**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

What is the variance of rolling a fair six-sided die? Student's answer: 3.5

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: 3.5 is the mean. The variance is $\frac{35}{12}$, about 2.92.

Correct answer: $\frac{35}{12}$, about 2.92

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A random variable has variance 9. What is its standard deviation? Student's answer: 9

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The standard deviation is $\sqrt{9} = 3$.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A ticket costs \$2. It wins \$6 with probability $\frac{1}{4}$ and loses otherwise. Is this a fair game? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Expected winnings are $6 \times \frac{1}{4} = 1.5$ and the cost is 2. The expected value is -0.5, so the game is not fair.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A game pays $10 with probability 0.1 and $0 otherwise. What is a fair price to play? Student's answer: $10

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The expected payout is 10 × 0.1 = $1, so a fair price is $1.**

Correct answer: $1

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A game pays \$5 with probability $\frac{1}{2}$ and \$0 otherwise. What is a fair price to play? Student's answer: \$5

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The expected payout is $5 \times \frac{1}{2} = \$2.50$.**

Correct answer: **\$2.50**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A product earns $100 with probability 0.6 and loses $50 with probability 0.4. What is the expected value? Student's answer: 60

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The student ignored the loss. E = 100(0.6) - 50(0.4) = 60 - 20 = 40.**

Correct answer: 40

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Option A has expected value 3 and option B has expected value 2.5. Which should be chosen to maximize expected value? Student's answer: B

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **3 is greater than 2.5, so option A has the greater expected value.**

Correct answer: **A**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

An insurance premium is $200. The chance of a $5,000 loss is 0.02. Is the premium lower than the expected loss? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **The expected loss is 5,000 × 0.02 = $100. The premium of $200 is higher than $100.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

A player wins if a fair die shows an even number and loses otherwise. Is this game fair? Student says: No

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **Even numbers are 3 out of 6, so each side has probability $\frac{1}{2}$. The game is fair.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Four students use a fair spinner with 4 equal sections to choose one student for a prize. What is each student's chance? Student's answer: $\frac{1}{3}$

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **There are 4 equal sections, so each student has a $\frac{1}{4}$ chance.**

Correct answer: $\frac{1}{4}$

Find the Error

Name: _____ Date: _____

Look at the student's work below. Decide whether it is right. If it is wrong, find the error and solve the problem correctly.

Two dice are rolled. A player wins if the sum is 7 or more, and the other player wins otherwise. Is this fair? Student says: Yes

1. Did the student get it correct or incorrect? If they did it incorrectly, what did they do wrong?

2. Solve the problem correctly. Show all of your work.

The error: **There are 21 outcomes with a sum of 7 or more out of 36, which is $\frac{7}{12}$, not $\frac{1}{2}$. The game is not fair.**

Correct answer: **No**